Tak Hur

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Google Scholar | GitHub | LinkedIn | Website

Experiences

University of Waterloo — Canada
Postdoctoral Researcher — *Sept, 2026 –*

École Polytechnique — France
Visiting Intern — *Oct, 2025 – May, 2026*

Education

Yonsei University — Korea
PhD in Statistics and Data Science — *March, 2023 – August, 2026*

Imperial College London — United Kingdom
Msci Physics with Theoretical Physics — *Sept, 2021 – August, 2022*

Imperial College London — United Kingdom
Bsc Physics with Theoretical Physics — *Sept, 2017 – August, 2021*

Publications

(* indicates first author)

1. **Hur, T.*** (2026). "Stochastic Reconfiguration as Spectral Filters for Overparameterized Neural Quantum States". *NeurIPS (2026)*
2. Lee, C. **Hur, T.***, Park, D. (2025). "Scalable Neural Decoders for Practical Real-Time Quantum Error Correction". *npj QI (2026)*
3. Kim, Y. Im, C. Kim, T. **Hur, T.** Park, D. (2025). "Multi-channel convolutional neural quantum embedding". *Advanced Quantum Technologies, e00575 (2025)*
4. Liu, H. **Hur, T.*** Zhang, S. et al. (2025). "Neural Quantum Embedding via Deterministic Quantum Computation with One Qubit". *Phys. Rev. Lett. 135, 080603 (2025)*
5. **Hur, T.*** Park, D. (2025). "Understanding Generalization in Quantum Machine Learning with Margins". *ICML (2025)*
6. Choi, J. **Hur, T.** Park, D. Shin N. Lee, S. Lee, H. Han, S. (2025). "Early-stage detection of cognitive impairment by hybrid quantum-classical algorithm using resting-state functional MRI time-series". *Knowledge-Based Systems 310 112922 (2025)*
7. **Hur, T.*** Araujo, I. Park, D. (2024). "Neural Quantum Embedding: Pushing the Limits of Quantum Supervised Learning". *Phys. Rev. A 110, 022411 (2024)*
8. Hahm, J. **Hur, T.*** Lee, J. Park, D. (2024). "Generative Modeling of Quantum Distribution with Functional Flow Matching". *QTML (2024)*
9. **Hur, T.*** Kim, L. Park, D. (2022). "Quantum convolutional neural network for classical data classification". *Quantum Machine Intelligence 4, 3 (2022)*

Preprints

1. Ryu, J. **Hur, T.*** Park, D. (2026). ”Margin-Based Generalization for Quantum Machine Learning with Classical Data Encoding”. *arXiv.2610.24426*
2. Lee, C. **Hur, T.** Jae, J. Park, D. (2026). ”Machine Learning Approaches to Decoding Topological Quantum Codes”. *arXiv.2608.15760*
3. Choi, J. **Hur, T**, Jeong, S. et al. (2025) ”Optimizing Quantum Data Embeddings for Ligand-Based Virtual Screening”. *arXiv.2512.16177*

Talks

1. ”Scalable Neural Decoders for Practical Real-Time Quantum Error Correction” contributed talk at *Machine Learning to Advance Quantum Technologies* (ML4QT). September, 2026
2. ”Scalable Neural Decoders for Practical Real-Time Quantum Error Correction” short talk at *Quantum Techniques in Machine Learning* (QTML). Nov, 2025
3. ”Understanding Generalization in Quantum Machine Learning with Margins” Invited talk at *Jens Eisert’s Group Seminar*. March, 2025
4. ”Understanding Generalization in Quantum Machine Learning with Margins” Invited talk at *2024 KIAS Quantum Information Workshop*. Aug, 2024
5. ”Neural Quantum Embedding” Invited talk at *IBM Quantum Information Techinal Exchange* (IBM QITE). Apr, 2024
6. ”Neural Quantum Embedding” short talk at *Quantum Techniques in Machine Learning* (QTML). Nov, 2023
7. ”Quantum Kernel Optimization via Deep Learning.” *Electronics and Telecommunications Research Institute* (ETRI). Jan, 2023

Poster Presentations

1. ”Stochastic Reconfiguration as Spectral Filters for Overparameterized Neural Quantum States” Poster presentation at *Neural Information Processing Systems* (NeurIPS). Dec, 2026
2. ”Understanding Generalization in Quantum Machine Learning with Margins” Poster presentation at *International Conference on Machine Learning* (ICML). July, 2025
3. ”Understanding Generalization in Quantum Machine Learning with Margins” Poster presentation at *Quantum Techniques in Machine Learning* (QTML). Nov, 2024
4. ”Generative Modeling of Quantum Distribution with Functional Flow Matching” Poster presentation at *Quantum Techniques in Machine Learning* (QTML). Nov, 2024
5. ”Neural Quantum Embedding” Poster presentation at *Asian Quantum Information Science Conference* (AQIS). Aug, 2023
6. ”Neural Quantum Embedding” Poster presentation at *Joint Symposium on Quantum Computing* (JSQC). Aug, 2023

Grants & Awards

- NVIDIA Academic Grant (2026). 30k H100 GPU hours for “Mixture-of-Experts Decoders for Large-Scale Real-Time Quantum Error Correction”
- NVIDIA Academic Grant (2025). 30k A100 GPU hours for “Scalable Neural Decoders for Practical Real-Time Quantum Error Correction”
- Quantum Education and Training Support Program Grant 2025 (20 nationwide, funded by the Ministry of Science and ICT, Republic of Korea)
- IBM Quantum Leadership Program 2023 (15 nationwide)
- Third Place – IBM Qiskit Hackathon Taiwan (2023)
- Top 4 – Pennylane QHACK Coding Challenge (2022, out of 800+ teams)
- Top 10 – Pennylane QHACK Coding Challenge (2021, out of 800+ teams)

Technical Skills

- **Quantum SDKs:** PennyLane, CUDA-Q, Qiskit
- **Machine Learning Frameworks:** PyTorch, JAX
- **Programming Languages:** Python, Julia

Professional Service

1. **Journal Reviewer:** *PRX Quantum*, *Physical Review E*, *Quantum Information Processing*, EPJ Quantum Technology
2. **Conference Reviewer:** *Neurips 2026, AISTATS 2027*
3. **Teaching Assistant:** Statistical Methods, Quantum Machine Learning (Yonsei University)

References

[1] Prof. Kyungdeock Park, dkd.park@yonsei.ac.kr

- *Associate Professor* at Department of Applied Statistics, Department of Statistics and Data Science, Yonsei University

[2] Dr. Maria Schuld, maria@xanadu.ai

- *Senior Researcher and Software Developer* at Xanadu Quantum Technologies Inc.

[3] Prof. Youngseok Kim, kim.10017@osu.edu

- *Assistant Professor* at Department of Electrical and Computer Engineering, Ohio State University